Water quality testing

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The input to the problem:

$$\mathcal{I} := \left\{ \begin{array}{rll} N := \{\text{laboratroies}\}, & |N| = n, \\ T := \{\text{contaminants}\}, & |T| = t, \\ M[i][j] := \{k \in T \mid \text{device } j \text{ from lab } i \text{ tests } k\}, & i \in N,\ j \in [m], \\ c[i][j] := \text{purchase cost of device } j \text{ from lab } i, & c[i][j] \geq 0, \\ d[i] := \text{delivery fee of laboratory } i, & d[i] \geq 0. \end{array} \right\}.$$

The output is a set of selected testing devices

$$\mathcal{S} := \{(i,j) \mid \text{we select device } j \text{ from laboratory } i\}.$$

$$\text{feasible}(\mathcal{S}) := \mathcal{S} \subseteq N \times [m] \ \wedge \bigcup_{(i,j)\in\mathcal{S}} M[i][j] = T.$$

The cost function is

$$f(\mathcal{S}) := \sum_{(i,j)\in\mathcal{S}} c[i][j] + \sum_{\substack{i\in N \\ \exists j\in[m]:\,(i,j)\in\mathcal{S}}} d[i].$$

Goal 1. For the natural LP given in the exercise, show that:

$$\forall B > 0,\ \exists \mathcal{I}: \quad |N| = |T| = 1 \ \wedge\ \mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}) > B\,\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}).$$

$\mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I})$: minimum cost of a feasible selection of devices,
$\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I})$: optimal value of the LP given in the exercise.

Fix an arbitrary $B > 0$. We construct an instance $\mathcal{I}_B$ depending on B, with one laboratory and one contamimant, such that

$$\mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}_B) > B\,\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}_B).$$

Choose one laboratory, one contaminant, and $m =?$ devices, each testing that contaminant. Set

$$d_1 = 1, \qquad c_{1j} = 0 \quad \forall j \in [m].$$

Since there is only one laboratory, the LP becomes

$$\begin{aligned} \text{minimize} \quad & d_1 y_1 + \sum_{j=1}^{m} c_{1j} x_{1j} = y_1 \\ \text{subject to} \quad & \sum_{j=1}^{m} x_{1j} \geq 1, \\ & x_{1j} \leq y_1 && \forall j \in [m], \\ & 0 \leq y_1, x_{1j} \leq 1 && \forall j \in [m]. \end{aligned}$$

To minimize y_1, we choose

$$x_{1j} = y_1 = \frac{1}{m} \qquad \forall j \in [m].$$

This is feasable. To see that a smaller cost is impossible, suppose $y_1 < 1/m$. Since $x_{1j} \leq y_1$ for every device,

$$\sum_{j=1}^{m} x_{1j} \leq m y_1 < 1,$$

contradicting the coverage constraint. Thus,

$$\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}_B) = \frac{1}{m}.$$

For the ILP, the variables are in $\{0, 1\}$. The coverage constraint requires at least one device with $x_{1j} = 1$, which forces $y_1 = 1$. Selecting just one device is feasible and costs 1, since all device costs are zero. Therefore,

$$\mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}_B) = 1.$$

Substituting these values into our target inequality gives

$$\mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}_B) > B\,\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}_B) \iff 1 > \frac{B}{m} \iff m > B.$$

Thus, choosing any integer $m > B$ proves our claim. □

Goal 2. Given the strengthened LP from the exercise, we have to show that:

$$\forall \mathcal{S} \text{ feasible for the ILP}, \quad \exists x_{ij},\ y_i,\ z_{ik} \quad (i \in N,\ j \in [m],\ k \in T):$$
$$\text{all constraints of the strengthened LP hold, and}$$
$$\sum_{i=1}^{n} d_i y_i + \sum_{i=1}^{n} \sum_{j=1}^{m} c_{ij} x_{ij} = f(\mathcal{S}).$$

The old constraints bound each device separately by $x_{ij} \leq y_i$. The new constraint bounds the laboratory's total contribution to contaminant k by $z_{ik} \leq y_i$.

Let us choose a generic feasible selection $\mathcal{S}$. For $i \in N$, $j \in [m]$, and $k \in T$, set

$$x_{ij} := \begin{cases} 1 & (i,j) \in \mathcal{S}, \\ 0 & \text{otherwise}, \end{cases}$$

$$y_i := \begin{cases} 1 & \exists j \in [m] :\ (i,j) \in \mathcal{S}, \\ 0 & \text{otherwise}, \end{cases}$$

$$z_{ik} := \begin{cases} 1 & \exists j \in [m] :\ (i,j) \in \mathcal{S} \ \wedge\ k \in M[i][j], \\ 0 & \text{otherwise}. \end{cases}$$

These variables satisfy the new constaints:

1. $z_{ik} \leq y_i$ holds because if $z_{ik} = 1$, a selected device from laboratory i tests k, so $y_i = 1$.
2. $z_{ik} \leq \sum_{j:\, k \in M[i][j]} x_{ij}$ holds because if $z_{ik} = 1$, at least one selected device appears in the sum, so the sum is at least 1.
3. $\sum_i z_{ik} \geq 1$ holds since $\mathcal{S}$ is feasible, so every contaminant k is tested by a selected device from some laboratory.

All variables are in $\{0,1\}$, selecting a device implies using its laboratory, and every contaminant is covered. Thus, the original LP constraints also hold.

Finally, the objective charges exactly the selected devices and one delivery fee per used laboratory, giving

$$\sum_{i=1}^{n} d_i y_i + \sum_{i=1}^{n}\sum_{j=1}^{m} c_{ij} x_{ij} = \sum_{\substack{i \in N \\ \exists j:\,(i,j)\in\mathcal{S}}} d[i] + \sum_{(i,j)\in\mathcal{S}} c[i][j] = f(\mathcal{S}).$$

Thus, for any feasible $\mathcal{S}$, we can choose the variables as above so that all constraints of the strengthened LP hold and the cost equals $f(\mathcal{S})$. □

Goal 3. Let $\mathcal{A}$ be our randomized procedure and let $\mathcal{S}_{\mathcal{A}}$ be the set of devices it selects. The exercise asks us to prove that:

1. Expected cost.
$$\mathbb{E}[f(\mathcal{S}_{\mathcal{A}})] \le \mathrm{OPT}_{\mathrm{LP}}.$$
2. Coverage probability.
$$\exists p > 0 \text{ constant}, \quad \forall k \in T: \quad \Pr\left[k \in \bigcup_{(i,j)\in\mathcal{S}_{\mathcal{A}}} M[i][j]\right] \ge p.$$

Here we use the usual trick from the lecture and use the fractional LP values as probabilities. Let x_{ij}, y_i, z_{ik} be an optimal solution of the strengthened LP. Select each laboratory i independently with probability y_i. If it is selected, select each of its devices j independently with probability x_{ij}/y_i. If $y_i = 0$, select no devices from that laboratory.

We divide by y_i to compensate for selecting the laboratory first. For $y_i > 0$, the overall probability of selecting device (i,j) is

$$\begin{aligned}\Pr[(i,j)\in\mathcal{S}_{\mathcal{A}}] &\overset{\text{multiplication rule}}{=} \Pr[\text{select laboratory } i] \\ &\quad \cdot \Pr[\text{select device } j \mid \text{laboratory } i \text{ selected}] \\ &= y_i \cdot \frac{x_{ij}}{y_i} = x_{ij}.\end{aligned}$$

If $y_i = 0$, then $x_{ij} = 0$ by $x_{ij} \le y_i$, so this equality still holds. All probabilities are valid because $0 \le x_{ij} \le y_i \le 1$.

Now we calculate the expected cost. Let Y_i indicate that laboratory i is selected, and let X_{ij} indicate that device (i,j) is selected. Then $\mathbb{E}[Y_i] = y_i$ and $\mathbb{E}[X_{ij}] = x_{ij}$. If $Y_i = 1$ but no device from laboratory i ends up selected, its actual delivery fee is 0. Therefore,

$$\text{actual delivery cost from laboratory } i \le d_i Y_i.$$

By linearity of expectation,

$$\begin{aligned}\mathbb{E}[f(\mathcal{S}_{\mathcal{A}})] &\le \mathbb{E}\left[\sum_{i=1}^{n} d_i Y_i + \sum_{i=1}^{n}\sum_{j=1}^{m} c_{ij} X_{ij}\right] \\ &= \sum_{i=1}^{n} d_i \mathbb{E}[Y_i] + \sum_{i=1}^{n}\sum_{j=1}^{m} c_{ij}\mathbb{E}[X_{ij}] \\ &= \sum_{i=1}^{n} d_i y_i + \sum_{i=1}^{n}\sum_{j=1}^{m} c_{ij} x_{ij} = \mathrm{OPT}_{\mathrm{LP}}.\end{aligned}$$

Now we analyze the coverage probability. Using the complement event and independence between laboratories,

$$\begin{aligned}\Pr\left[k \in \bigcup_{(i,j)\in\mathcal{S}_{\mathcal{A}}} M[i][j]\right] &= 1 - \Pr[k \text{ is not tested}] \\ &= 1 - \prod_{i\in N} \Pr[\text{laboratory } i \text{ does not test } k] \\ &= 1 - \prod_{i\in N} \big(1 - \Pr[\text{laboratory } i \text{ tests } k]\big).\end{aligned}$$

For each laboratory with $y_i > 0$, the multiplication rule gives

$$\begin{aligned}\Pr[\text{laboratory } i \text{ tests } k] = \Pr[&\text{select laboratory } i] \\ &\cdot \Pr[\text{laboratory } i \text{ tests } k \mid \text{laboratory } i \text{ selected}].\end{aligned}$$

$$\begin{aligned}&\Pr[\text{laboratory } i \text{ tests } k \mid \text{laboratory } i \text{ selected}] \\ &\quad = 1 - \Pr[\text{no device testing } k \text{ selected} \mid \text{laboratory } i \text{ selected}] \\ &\quad = 1 - \prod_{\substack{j\in[m] \\ k\in M[i][j]}} \Pr[\text{device } (i,j) \text{ not selected} \mid \text{laboratory } i \text{ selected}] \\ &\quad = 1 - \prod_{\substack{j\in[m] \\ k\in M[i][j]}} \left(1 - \Pr[\text{select device } (i,j) \mid \text{laboratory } i \text{ selected}]\right).\end{aligned}$$

Substituting the probabilities gives

$$\begin{aligned}\Pr\left[k \in \bigcup_{(i,j)\in\mathcal{S}_{\mathcal{A}}} M[i][j]\right] &= 1 - \prod_{\substack{i\in N \\ y_i>0}} \left[1 - y_i \left(1 - \prod_{\substack{j\in[m] \\ k\in M[i][j]}} \left(1 - \frac{x_{ij}}{y_i}\right)\right)\right] \\ &\overset{(1)}{\geq} 1 - \prod_{\substack{i\in N \\ y_i>0}} \left[1 - y_i \left(1 - \exp\left(- \sum_{\substack{j\in[m] \\ k\in M[i][j]}} \frac{x_{ij}}{y_i}\right)\right)\right] \\ &\overset{(2)}{\geq} 1 - \prod_{\substack{i\in N \\ y_i>0}} \left[1 - y_i \left(1 - e^{-z_{ik}/y_i}\right)\right] \\ &\overset{(3)}{\geq} 1 - \prod_{\substack{i\in N \\ y_i>0}} \left[1 - (1 - e^{-1}) z_{ik}\right] \\ &\overset{(4)}{\geq} 1 - \exp\left(-(1 - e^{-1}) \sum_{\substack{i\in N \\ y_i>0}} z_{ik}\right) \\ &\overset{(5)}{\geq} 1 - e^{-(1-e^{-1})}.\end{aligned}$$

Laboratories with $y_i = 0$ are omitted because they are never selected. The numbered steps use the following facts.

1. As in many exercises, the factors $1 - u$ suggest using the familiar bound $1 - u \leq e^{-u}$ for $0 \leq u \leq 1$. Since $0 \leq x_{ij}/y_i \leq 1$, we apply it to each factor in the inner product. Multiplying the exponential factors gives the exponential of their sum.

2. The LP constraint gives $z_{ik} \le \sum_{j:\, k \in M[i][j]} x_{ij}$, so $e^{-\sum_j x_{ij}/y_i} \le e^{-z_{ik}/y_i}$, where the sum is over devices testing k.
3. By $0 \le z_{ik} \le y_i$, the ratio $u = z_{ik}/y_i$ lies in $[0,1]$. The function $h(u) = 1 - e^{-u}$ is concave on $[0,1]$ because $h''(u) = -e^{-u} < 0$ for every $u \in [0,1]$. As we have done many times in the exercises, we use Jensen's inequality for concave functions,
$$1 - e^{-u} = h((1-u)\cdot 0 + u \cdot 1) \ge (1-u)h(0) + uh(1) = (1 - e^{-1})u.$$
Multiplying by y_i gives $y_i(1 - e^{-z_{ik}/y_i}) \ge (1 - e^{-1})z_{ik}$.
4. Apply $1 - u \le e^{-u}$ again, now with $u = (1 - e^{-1})z_{ik}$, to the outer product. Since $0 \le z_{ik} \le y_i \le 1$ and $0 < 1 - e^{-1} < 1$,
$$0 \le (1 - e^{-1})z_{ik} \le 1 - e^{-1} < 1,$$
so $u \in [0,1]$.
5. If $y_i = 0$, then $z_{ik} = 0$. Hence the remaining sum equals $\sum_i z_{ik} \ge 1$ by the coverage constraint of the strengthened LP.

Thus, we can take the constant
$$p = 1 - e^{-(1-e^{-1})} \approx 0.4685 > 0,$$
which is independent of the input. □

Goal 4. Design a polynomial-time randomized algorithm $\mathcal{A}$ such that
$$\Pr[\text{feasible}(\mathcal{S}_{\mathcal{A}}) \ \wedge \ f(\mathcal{S}_{\mathcal{A}}) \le O(\log n)\,\mathrm{OPT}_{\mathrm{ILP}}] \ge 1 - \delta \qquad \text{for fixed } \delta \in (0,1).$$
The procedure from Goal 3 gives a selected set of devices, but that set need not be feasible, since it may not cover all contaminants. The idea is therefore to run the procedure several times and combine the selected devices. Call the procedure from Goal 3 $\mathcal{A}_3$ and its output $\mathcal{S}_{\mathcal{A}_3}$. Run it independently r times and let
$$\mathcal{S} := \bigcup_{\ell=1}^{r} \mathcal{S}_{\mathcal{A}_3}^{(\ell)}.$$
We first separate the two possible failures using the complement rule and the union bound,
$$\begin{aligned} &\Pr[\text{feasible}(\mathcal{S}) \ \wedge \ f(\mathcal{S}) \le O(\log n)\,\mathrm{OPT}_{\mathrm{ILP}}] \\ &\quad = 1 - \Pr[\neg\,\text{feasible}(\mathcal{S}) \ \vee \ f(\mathcal{S}) > O(\log n)\,\mathrm{OPT}_{\mathrm{ILP}}] \qquad (*) \\ &\quad \ge 1 - \Pr[\neg\,\text{feasible}(\mathcal{S})] - \Pr[f(\mathcal{S}) > O(\log n)\,\mathrm{OPT}_{\mathrm{ILP}}]\,. \end{aligned}$$
Let us first find $\Pr[\neg\,\text{feasible}(\mathcal{S})]$.
$$\begin{aligned} \Pr[\neg\,\text{feasible}(\mathcal{S})] &= \Pr[\text{some contaminant is missed in all } r \text{ rounds}] \\ &= \Pr\left[\bigcup_{k \in T} \{k \text{ is missed in all } r \text{ rounds}\}\right] \\ &\le \sum_{k \in T} \Pr[k \text{ is missed in all } r \text{ rounds}]. \end{aligned}$$
For each contaminant $k \in T$, independence of the runs gives
$$\begin{aligned} \Pr[k \text{ is missed in all } r \text{ rounds}] &= \prod_{\ell=1}^{r} \Pr[k \text{ is missed in round } \ell] \\ &= \prod_{\ell=1}^{r} (1 - \Pr[k \text{ is tested in round } \ell]) \\ &\le (1-p)^r, \end{aligned}$$

where p is the constant from Goal 3. Therefore,

$$\Pr[\neg\,\text{feasible}(\mathcal{S})] \le \sum_{k\in T}(1-p)^r = t(1-p)^r.$$

What remains is to bound

$$\Pr[f(\mathcal{S}) > O(\log n)\,\mathrm{OPT}_{\mathrm{ILP}}]\,.$$

We have not yet determined the factor represented by $O(\log n)$, so write $a =?$ for this positive factor. This probability has the form needed for Markov's inequality. For $\mathrm{OPT}_{\mathrm{ILP}} > 0$,

$$\Pr[f(\mathcal{S}) > a\,\mathrm{OPT}_{\mathrm{ILP}}] \overset{\text{Markov}}{\le} \frac{\mathbb{E}[f(\mathcal{S})]}{a\,\mathrm{OPT}_{\mathrm{ILP}}}.$$

To continue, we need to bound $\mathbb{E}[f(\mathcal{S})]$.

$$\begin{aligned}
\mathbb{E}[f(\mathcal{S})] &= \mathbb{E}\left[f\left(\bigcup_{\ell=1}^{r}\mathcal{S}^{(\ell)}_{\mathcal{A}_3}\right)\right] \\
&\le \mathbb{E}\left[\sum_{\ell=1}^{r} f(\mathcal{S}^{(\ell)}_{\mathcal{A}_3})\right] \\
&\overset{\text{linearity}}{=} \sum_{\ell=1}^{r}\mathbb{E}[f(\mathcal{S}^{(\ell)}_{\mathcal{A}_3})] \overset{\substack{\text{expected cost}\\ \text{in Goal 3}}}{\le} r\,\mathrm{OPT}_{\mathrm{LP}} \le r\,\mathrm{OPT}_{\mathrm{ILP}}.
\end{aligned}$$

Substituting this expected-cost bound gives

$$\Pr[f(\mathcal{S}) > a\,\mathrm{OPT}_{\mathrm{ILP}}] \le \frac{r\,\mathrm{OPT}_{\mathrm{ILP}}}{a\,\mathrm{OPT}_{\mathrm{ILP}}} = \frac{r}{a}.$$

If $\mathrm{OPT}_{\mathrm{ILP}} = 0$, then

$$0 \le \mathbb{E}[f(\mathcal{S})] \le r\,\mathrm{OPT}_{\mathrm{ILP}} = 0.$$

Since $f(\mathcal{S}) \ge 0$,

$$\mathbb{E}[f(\mathcal{S})] = 0 \implies \Pr[f(\mathcal{S}) = 0] = 1.$$

Therefore,

$$\Pr[f(\mathcal{S}) > a\,\mathrm{OPT}_{\mathrm{ILP}}] = \Pr[f(\mathcal{S}) > 0] = 0 \le \frac{r}{a}.$$

We will choose a and r later, and must ensure $a = O(\log n)$. Now we substitute both probability bounds into Equation (*), writing the unknown cost factor as a, to obtain

$$\Pr[\text{feasible}(\mathcal{S}) \ \wedge\ f(\mathcal{S}) \le a\,\mathrm{OPT}_{\mathrm{ILP}}] \ge 1 - t(1-p)^r - \frac{r}{a}.$$

To make this at least $1-\delta$, we need

$$t(1-p)^r + \frac{r}{a} \le \delta.$$

We give each failure probability half of the allowed error by requiring

$$t(1-p)^r \le \frac{\delta}{2}, \qquad \frac{r}{a} \le \frac{\delta}{2}.$$

Together, these imply $t(1-p)^r + r/a \le \delta$. Choose

$$r = \left\lceil \frac{\ln(2t/\delta)}{-\ln(1-p)} \right\rceil, \qquad a = \frac{2r}{\delta}.$$

Substituting these choices gives

$$\Pr\left[\text{feasible}(\mathcal{S}) \ \wedge \ f(\mathcal{S}) \leq \frac{2r}{\delta}\,\mathrm{OPT}_{\mathrm{ILP}}\right] \geq 1 - \frac{\delta}{2} - \frac{\delta}{2} = 1 - \delta.$$

For fixed δ, this gives an $O(\log t)$ approximmation. The exercise asks for an $O(\log n)$ approximmation, but no relation between t and n is given here. I could not find an obvious way to change this algorithm so that the bound depends on n instead of t. For fixed δ, the number of runs is $r = O(\log t)$. Since each run of $\mathcal{A}_3$ takes polynomial time, the whole procedure also takes polynomial time.